BEE 4750/5750

Homework 6: Linear Programs and Shadow Prices

2026-10-21

Due Date
Thursday, 10/29/26, 9:00pm

Overview

Instructions

- Problem 1 asks you to formulate a water-supply problem as a linear program.
- Problem 2 asks you to interpret a linear program graphically.
- Problem 3 asks you to find the optimal solution to a stormwater plan and its shadow prices by hand, and to see where a shadow price stops being a good estimate.
- Problem 4 is a larger nutrient-management LP. The `JuMP` code that solves it is provided; you interpret its solution.
- Write your answers in Markdown cells. For the sketches, a photo or scan of a hand drawing is fine: save it in this folder and insert it with `![](sketch.png)`.

Load Environment

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
using JuMP
using HiGHS
using Plots
```

Problems (Total: 50 Points)

Problem 1 (13)

A city of about 150,000 people supplies its drinking water from two sources: a river intake and a groundwater well field. It must deliver at least 20 million gallons per day (MGD). The operating cost (energy and chemicals) of treating and pumping river water is \$300 per million gallons;

for groundwater, which has to be pumped from deeper, it is $500 per million gallons. The river withdrawal permit allows at most 15 MGD, and the well field can produce at most 12 MGD.

River water carries 8 mg/L of nitrate from farms upstream; the groundwater carries 2 mg/L. To stay well below the 10 mg/L drinking-water standard, the city requires the blended supply to average at most 6 mg/L of nitrate. The city wants to meet demand at the lowest daily cost.

Problem 1.1 (1)
Define the decision variables, with units.

Problem 1.2 (2)
What are the bounds on each decision variable, and where does each bound come from?

Problem 1.3 (2)
Write the objective function, with units. Is the city maximizing or minimizing it?

Problem 1.4 (5)
Write every constraint in linear form, including the bounds from Problem 1.2, each with a short label saying what it represents and with units that agree on both sides.

Problem 1.5 (3)
Write the problem in standard form,

$$\max_{\mathbf{x}} \quad \mathbf{c}^\top \mathbf{x} \qquad \text{subject to} \quad \mathbf{A}\mathbf{x} \le \mathbf{b}, \quad \mathbf{x} \ge \mathbf{0},$$

where $\mathbf{x}$ is the vector of decision variables. $\mathbf{A}\mathbf{x} \le \mathbf{b}$ collects every constraint except $\mathbf{x} \ge \mathbf{0}$: $\mathbf{A}$ has one row, and $\mathbf{b}$ one entry, for each. Give the vector $\mathbf{c}$, the matrix $\mathbf{A}$, and the vector $\mathbf{b}$ explicitly.

Problem 2 (7)

Consider the linear program

$$\begin{aligned} \max_{x_1, x_2} \quad & z = 3x_1 + 2x_2 \\ \text{subject to:} \quad & x_1 + x_2 \le 8 \\ & 2x_1 + x_2 \le 12 \\ & x_2 \le 6 \\ & x_1, x_2 \ge 0. \end{aligned}$$

Problem 2.1 (2)
Sketch the feasible region. Label each constraint's boundary line, and label every corner of the region with its coordinates.

Problem 2.2 (5)
Say whether each of the following points is optimal for this linear program, and justify each answer.

1. $(4, 4)$

2. $(2, 6)$
3. $(3, 5)$
4. $(2, 2)$
5. $(5, 3)$

Problem 3 (18)

A lakeside city is under a consent order to reduce the stormwater reaching its combined sewers and the phosphorus reaching its lake. It plans to build two kinds of green infrastructure in one sewershed:

- **Bioretention cells** (rain gardens): shallow planted basins that collect runoff from the surrounding pavement and roofs and filter it through soil. Each cell drains an area several times its own size.
- **Permeable pavement**: parking-lot surfaces that let rain soak through into a stone reservoir underneath. It mostly handles the rain that falls on it.

Per acre built, the two options have the following performance and cost characteristics. The costs are annualized: construction spread over the lifetime of the infrastructure plus maintenance.

Per acre built	Bioretention	Permeable pavement
Runoff captured (million gallons/yr)	5	2
Phosphorus removed (kg/yr)	5	1
Annualized cost ($1,000/yr)	50	30

The consent order requires the city to capture at least 40 million gallons of runoff per year and to remove at least 30 kg of phosphorus per year. Suitable sites exist for at most 6 acres of bioretention, and there are 15 acres of parking lots that could take permeable pavement. With B the acres of bioretention and P the acres of permeable pavement, the city's problem is

$$
\begin{aligned}
\min_{B,P} \quad & 50B + 30P && (\$1\ ,\ 000/\ \text{yr}) \\
\text{subject to:} \quad & 5B + 2P \geq 40 && \text{runoff captured (million gal/yr)} \\
& 5B + P \geq 30 && \text{phosphorus removed (kg/yr)} \\
& B \leq 6 && \text{bioretention sites (acres)} \\
& P \leq 15 && \text{parking lots (acres)} \\
& B, P \geq 0.
\end{aligned}
$$

Problem 3.1 (2)

Sketch the feasible region, with B on the horizontal axis, and label every corner with its coordinates.

Problem 3.2 (2)
Find the optimal plan, using level lines of the cost or by evaluating the cost at each corner, and report the minimum annual cost.

Problem 3.3 (1)
Which constraints are binding at the optimum?

Problem 3.4 (4)
Find the shadow prices of the two binding constraints, with units. One way: relax each binding constraint by one unit, find where the optimal corner moves, and compute how much the cost falls. You can check your answer with the method from lecture, writing the cost coefficients as a combination of the binding constraints' coefficients.

Problem 3.5 (1)
The shadow price of the phosphorus constraint is zero. What does that mean for the city?

Problem 3.6 (2)
A school district offers a 4-acre site for bioretention, but only as a single lease, at $60,000 per year for the whole site. Using only the shadow price of the bioretention-site constraint, estimate how much 4 more acres would save the city each year. On that estimate, should the city sign the lease?

Problem 3.7 (3)
Raise the bioretention limit from 6 to 10 acres. Find the new optimal plan and its annual cost by hand, by extending your sketch from Problem 3.1.

Problem 3.8 (1)
How much does the city actually save with the 10-acre limit, and should it sign the lease?

Problem 3.9 (2)
Look at your plan from Problem 3.7. Why is the shadow-price estimate from Problem 3.6 wrong for a change this large?

Problem 4 (12)
Nitrogen from farms and wastewater treatment plants feeds algal blooms in rivers, reservoirs and estuaries. A state agency is setting limits on the nitrogen load reaching three points on a river: a drinking-water **intake**, a **reservoir**, and the **estuary** at the river's mouth. Six sources release nitrogen to the river. Going downstream, the river passes the upper farms and an upstream town's treatment plant, then the intake; then the middle farms and a city's treatment plant, then the reservoir; then the lower farms and a coastal town's treatment plant, and then it reaches the estuary.

Not all the nitrogen a source releases reaches each point downstream of it: plants and algae take some up, bacteria return some to the air, and the reservoir traps about half of what flows into it. A source's **delivery ratio** to a point is the fraction of its nitrogen that reaches that point; it is 0 for a point upstream of the source. Each source can cut its nitrogen, with cover crops and buffer

strips on the farms or upgrades at the treatment plants, up to a maximum and at a cost per kg removed. Loads and removals are in kg of nitrogen per day:

Source	Load with no cuts	Maximum removal	Cost per kg removed
Upper farms	1,000	300	$3.50
Upstream town's plant	200	160	$5
Middle farms	1,200	360	$4
City's plant	1,500	1,200	$8
Lower farms	800	240	$6
Coastal town's plant	300	240	$7

Delivery ratio	Intake	Reservoir	Estuary
Upper farms	0.4	0.3	0.15
Upstream town's plant	0.8	0.6	0.3
Middle farms	0	0.6	0.3
City's plant	0	0.8	0.4
Lower farms	0	0	0.6
Coastal town's plant	0	0	0.9

The limits are 500 kg/day at the intake, 1,500 kg/day at the reservoir and 1,200 kg/day at the estuary. The agency wants the cheapest set of cuts that meets all three.

With six decision variables, you can't draw the feasible region, and it has far too many corners to check by hand. The code below writes the problem as a linear program in `JuMP` and solves it; you only need to run it. The shadow prices then do the work the picture did in Problems 2 and 3: they explain why the optimal plan looks the way it does.

```
source_name = ["Upper farms", "Upstream town", "Middle farms",
    "City", "Lower farms", "Coastal town"]
nitrogen_load = [1000, 200, 1200, 1500, 800, 300]   # [kg/day] with no cuts
max_removal = [300, 160, 360, 1200, 240, 240]       # [kg/day]
removal_cost = [3.5, 5, 4, 8, 6, 7]                 # [$ per kg removed]

point_name = ["Intake", "Reservoir", "Estuary"]
load_limit = [500, 1500, 1200]                      # [kg/day]
# delivery[j, i]: the fraction of source i's nitrogen that reaches point j
delivery = [0.4  0.8  0    0    0    0;
            0.3  0.6  0.6  0.8  0    0;
            0.15 0.3  0.3  0.4  0.6  0.9]
```

```
n_sources = length(nitrogen_load)
n_points = length(load_limit)

nutrients = Model(HiGHS.Optimizer)
set_silent(nutrients)

# removal at each source [kg/day], up to its maximum
@variable(nutrients, 0 <= removal[i in 1:n_sources] <= max_removal[i])
# daily cost [$/day]
@objective(nutrients, Min,
    sum(removal_cost[i] * removal[i] for i in 1:n_sources))
# the nitrogen reaching each point [kg/day] stays within its limit
@constraint(nutrients, limit[j in 1:n_points],
    sum(delivery[j, i] * (nitrogen_load[i] - removal[i])
        for i in 1:n_sources) <= load_limit[j])

optimize!(nutrients)
println("Daily cost: \$", round(objective_value(nutrients); digits=2))

# How much each source removes [kg/day]. Adding 0 after rounding
# turns a -0.0 from the solver into 0
for i in 1:n_sources
    kg_removed = round(value(removal[i]); digits=1) + 0
    println(source_name[i], ": removes ", kg_removed, " of at most ",
        max_removal[i], " kg/day")
end

# The nitrogen reaching each point with the plan [kg/day], and the
# shadow price of the point's limit
reaching = zeros(n_points)
for j in 1:n_points
    for i in 1:n_sources
        kg_left = nitrogen_load[i] - value(removal[i])
        reaching[j] = reaching[j] + delivery[j, i] * kg_left
    end
    limit_price = round(shadow_price(limit[j]); digits=2) + 0
    println(point_name[j], ": ", round(reaching[j]; digits=1),
        " kg/day reaching, limit ", load_limit[j],
        " kg/day, shadow price ", limit_price)
end
```

Problem 4.1 (2)
Run the model. Report the daily cost, how much each source removes (as a table), and the load reaching each of the three points, compared with its limit.

Problem 4.2 (3)
Which limits bind? Report the shadow price of each limit, as a positive number with units.

Problem 4.3 (4)
Write d_{ji} for source i's delivery ratio to point j and c_i for its cost per kg removed. Removing 1 more kg/day at source i lowers the load reaching each point j by d_{ji} kg/day. With λ_j the shadow price of point j's limit (as a positive number), that kilogram is worth

$$v_i = \sum_j \lambda_j d_{ji} \qquad (\$ \text{ per kg removed}).$$

Compute v_i for each source and compare it with the source's cost, c_i. Use the comparison to explain why each source removes nothing, less than its maximum, or its maximum.

Problem 4.4 (3)
The upper farms are the cheapest source per kg removed, and the city's plant is the most expensive. Why does the plan cut nothing at the upper farms, but more at the city's plant than at any other source?